

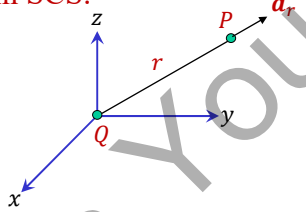
Electric Field Intensity (E)

$$\mathbf{F}_t = \frac{Q_1 Q_t}{4\pi\epsilon_o R_{1t}^2} \mathbf{a}_{1t}$$

$$\mathbf{E}_1 = \lim_{Q_t \rightarrow 0} \frac{\mathbf{F}_t}{Q_t}$$
$$\mathbf{E}_1 = \frac{Q_1}{4\pi\epsilon_o R_{1t}^2} \mathbf{a}_{1t} \left[\frac{N}{C} \right] \text{ or } \left[\frac{V}{m} \right]$$

Electric Field due to Point Charge ' Q ':
$$\mathbf{E} = \frac{Q}{4\pi\epsilon_o R^2} \mathbf{a}_R$$

Electric Field due to Point Charge ' Q ' in SCS:
Case 1: Point Charge ' Q ' at the Origin:
$$\mathbf{E} = \frac{Q}{4\pi\epsilon_o r^2} \mathbf{a}_r$$

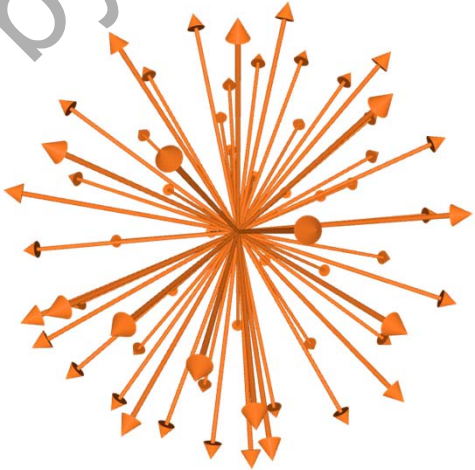


Thus, the point charge at the origin possesses spherical symmetry

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6

Field due to Point Charge ' Q ': Streamlines



<https://imagnetostl.com/view-wrl-online#convert>

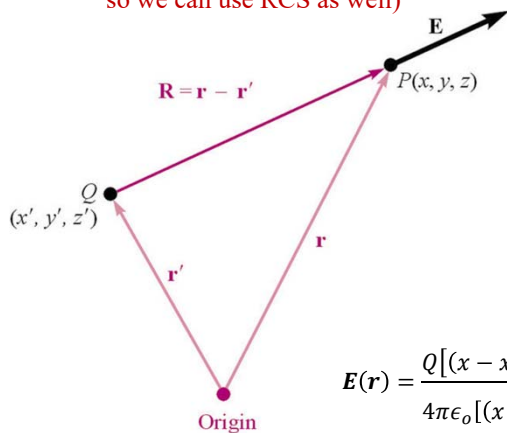
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7

Electric Field Intensity ($\mathbf{E}$)

Electric Field due to Point Charge ' Q ' in RCS:

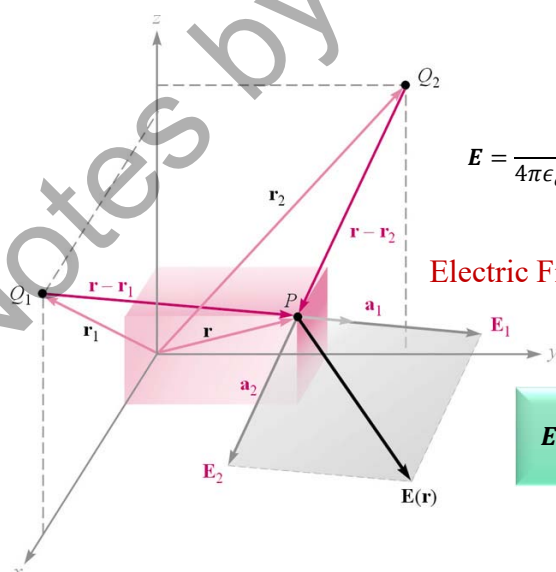
Case 2: Point Charge ' Q ' not at the Origin: (No spherical symmetry now, so we can use RCS as well)



$$\mathbf{r}' = \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = x' \mathbf{a}_x + y' \mathbf{a}_y + z' \mathbf{a}_z$$
$$\mathbf{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = x \mathbf{a}_x + y \mathbf{a}_y + z \mathbf{a}_z$$
$$\mathbf{R} = \mathbf{r} - \mathbf{r}'$$
$$\mathbf{E}(\mathbf{r}) = \frac{Q}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^2} \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|}$$
$$\mathbf{E}(\mathbf{r}) = \frac{Q(\mathbf{r} - \mathbf{r}')}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^3}$$
$$\mathbf{E}(\mathbf{r}) = \frac{Q[(x - x')\mathbf{a}_x + (y - y')\mathbf{a}_y + (z - z')\mathbf{a}_z]}{4\pi\epsilon_0 [(x - x')^2 + (y - y')^2 + (z - z')^2]^{\frac{3}{2}}}$$

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Electric Field Intensity ($\mathbf{E}$)



$$\mathbf{E}(\mathbf{r}) = \mathbf{E}_1 + \mathbf{E}_2$$
$$\mathbf{E} = \frac{Q_1}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_1|^2} \mathbf{a}_1 + \frac{Q_2}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_2|^2} \mathbf{a}_2$$

Electric Field due to ' n ' point charges:

$$\mathbf{E} = \sum_{i=1}^n \frac{Q_i}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_i|^2} \mathbf{a}_i$$

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Electric Field Intensity – Example 2.2

$$\mathbf{E} = 26.96 \left[\frac{\mathbf{a}_z}{1} \frac{1}{1^2} + \frac{2\mathbf{a}_x + \mathbf{a}_z}{\sqrt{5}} \frac{1}{(\sqrt{5})^2} + \frac{2\mathbf{a}_x + 2\mathbf{a}_y + \mathbf{a}_z}{3} \frac{1}{3^2} + \frac{2\mathbf{a}_y + \mathbf{a}_z}{\sqrt{5}} \frac{1}{(\sqrt{5})^2} \right]$$

$$\mathbf{E} = 6.82\mathbf{a}_x + 6.82\mathbf{a}_y + 32.8\mathbf{a}_z \text{ V/m}$$

D2.2. A charge of $-0.3 \mu\text{C}$ is located at $A(25, -30, 15)$ (in cm), and a second charge of $0.5 \mu\text{C}$ is at $B(-10, 8, 12)$ cm. Find $\mathbf{E}$ at: (a) the origin; (b) $P(15, 20, 50)$ cm.

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10

Continuous Charge Distributions as Point Charges

Point charge

Line charge

Surface charge

Volume charge

Point Charge ' Q ' [C]
 Line Charge Density ' ρ_L ' [C/m]
 $\rho_L = \lim_{dL \rightarrow 0} dQ/dL$
 Surface Charge Density ' ρ_S ' [C/m²]
 $\rho_S = \lim_{dS \rightarrow 0} dQ/dS$
 Volume Charge Density ' ρ_v ' [C/m³]
 $\rho_v = \lim_{dv \rightarrow 0} dQ/dv$

Finding Point Charge on a Line
 $dQ = \rho_L dL$ and total $Q = \int_L \rho_L dL$
 Finding Point Charge on a Surface
 $dQ = \rho_S dS$ and total $Q = \int_S \rho_S dS = \iint \rho_S dS$
 Find Point Charge in a Volume Charge
 $dQ = \rho_v dv$ and total $Q = \int_v \rho_v dv = \iiint \rho_v dv$

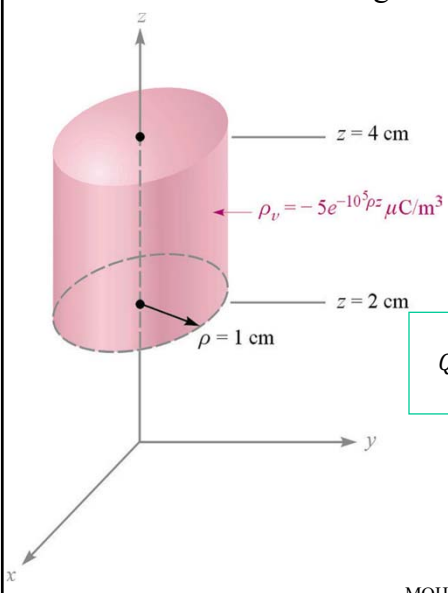
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11

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3

Continuous Volume Charge Distribution



Example 2.3
Find the total charge contained in a 2-cm length of electron beam as shown in Fig.

$$Q = \int_v \rho_v dv$$

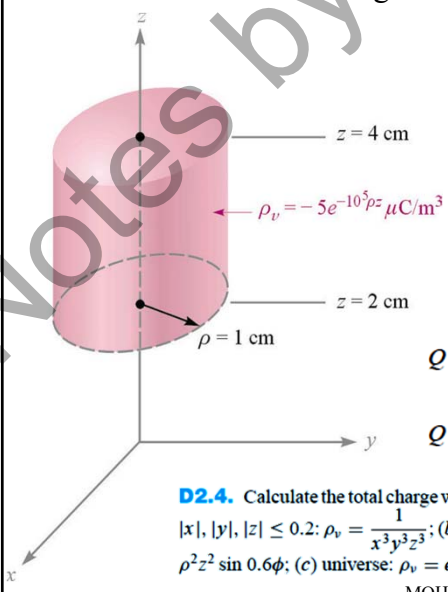
$$dv = \rho d\rho d\phi dz \text{ in CCS}$$

$$Q = \int_{z=0.02}^{0.04} \int_{\phi=0}^{2\pi} \int_{\rho=0}^{0.01} -5 \times 10^{-6} e^{-10^5 \rho z} \rho d\rho d\phi dz$$

$$Q = \int_{0.02}^{0.04} \int_0^{0.01} -10^{-5} \pi e^{-10^5 \rho z} \rho d\rho dz$$

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Continuous Volume Charge Distribution



Example 2.3
Find the total charge contained in a 2-cm length of electron beam as shown in Fig.

$$Q = \int_0^{0.01} \left(\frac{-10^{-5} \pi}{-10^5 \rho} e^{-10^5 \rho z} \rho d\rho \right)_{z=0.02}^{z=0.04}$$

$$= \int_0^{0.01} -10^{-5} \pi (e^{-2000 \rho} - e^{-4000 \rho}) d\rho$$

$$Q = -10^{-10} \pi \left(\frac{e^{-2000 \rho}}{-2000} - \frac{e^{-4000 \rho}}{-4000} \right)_0^{0.01}$$

$$Q = -10^{-10} \pi \left(\frac{1}{2000} - \frac{1}{4000} \right) = \frac{-\pi}{40} = 0.0785 \text{ pC}$$

D2.4. Calculate the total charge within each of the indicated volumes: (a) $0.1 \leq |x|, |y|, |z| \leq 0.2$; $\rho_v = \frac{1}{x^3 y^3 z^3}$; (b) $0 \leq \rho \leq 0.1, 0 \leq \phi \leq \pi, 2 \leq z \leq 4$; $\rho_v = \rho^2 z^2 \sin 0.6 \phi$; (c) universe: $\rho_v = e^{-2r}/r^2$.

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Field Due To Continuous Charge Distributions

$E = \frac{Q}{4\pi\epsilon_0 R^2} \mathbf{a}_R$



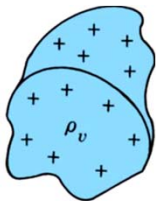
Point charge



Line charge



Surface charge



Volume charge

Field due to Line Charge

$$Q = \int_L \rho_L dL, \quad \mathbf{E} = \int_L \frac{\rho_L dL}{4\pi\epsilon_0 R^2} \mathbf{a}_R$$

Field due to Surface Charge

$$Q = \int_S \rho_S dS = \iint \rho_S dS, \quad \mathbf{E} = \int_S \frac{\rho_S dS}{4\pi\epsilon_0 R^2} \mathbf{a}_R$$

Field due to Volume Charge

$$Q = \int_v \rho_v dv = \iiint \rho_v dv, \quad \mathbf{E} = \int_v \frac{\rho_v dv}{4\pi\epsilon_0 R^2} \mathbf{a}_R$$

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14

Field Due To Uniform Line Charge Distribution

$E(\mathbf{r}) = \frac{Q}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^2} \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|}$



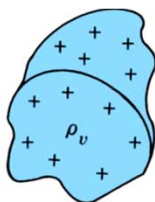
Point charge



Line charge



Surface charge



Volume charge

Field due to Line Charge with incremental point charge dQ at $\mathbf{r}'$

$$d\mathbf{E}(\mathbf{r}) = \frac{dQ}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^2} \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} = \frac{\rho_L dL}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^2} \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|}$$
$$\mathbf{E}(\mathbf{r}) = \int d\mathbf{E}(\mathbf{r}) = \int_L \frac{\rho_L dL}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^2} \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|}$$

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15

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5

Field Due To Uniform Surface Charge Distribution

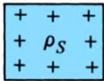
$$E(\mathbf{r}) = \frac{Q}{4\pi\epsilon_o|\mathbf{r}-\mathbf{r}'|^2} \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|}$$



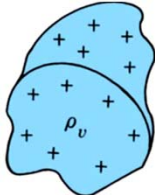
Point charge



Line charge



Surface charge



Volume charge

Field due to Surface Charge with incremental point charge dQ at $\mathbf{r}'$

$$dE(\mathbf{r}) = \frac{dQ}{4\pi\epsilon_o|\mathbf{r}-\mathbf{r}'|^2} \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|} = \frac{\rho_S dS}{4\pi\epsilon_o|\mathbf{r}-\mathbf{r}'|^2} \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|}$$
$$E(\mathbf{r}) = \int dE(\mathbf{r}) = \int_S \frac{\rho_S dS}{4\pi\epsilon_o|\mathbf{r}-\mathbf{r}'|^2} \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|}$$

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16

Field Due To Uniform Volume Charge Distribution

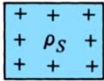
$$E(\mathbf{r}) = \frac{Q}{4\pi\epsilon_o|\mathbf{r}-\mathbf{r}'|^2} \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|}$$



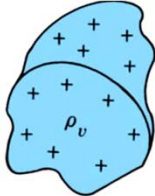
Point charge



Line charge



Surface charge



Volume charge

Field due to Volume Charge with incremental point charge dQ at $\mathbf{r}'$

$$dE(\mathbf{r}) = \frac{dQ}{4\pi\epsilon_o|\mathbf{r}-\mathbf{r}'|^2} \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|} = \frac{\rho_v dv}{4\pi\epsilon_o|\mathbf{r}-\mathbf{r}'|^2} \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|}$$
$$E(\mathbf{r}) = \int dE(\mathbf{r}) = \int_{vol} \frac{\rho_v dv}{4\pi\epsilon_o|\mathbf{r}-\mathbf{r}'|^2} \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|}$$

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17